

MOMENTS PARAMETERIZATION

CANONICAL PARAMETERIZATION

the *moments parameterization*. This is because the mean and covariance are the first and second moments of a probability distribution; all other moments are zero for normal distributions. In this chapter, we will also discuss an alternative parameterization, called *canonical parameterization*, or sometimes *natural parameterization*. Both parameterizations, the moments and the canonical parameterizations, are functionally equivalent in that a bijective mapping exists that transforms one into the other. However, they lead to filter algorithms with somewhat different computational characteristics. As we shall see, the canonical and the natural parameterizations are best thought of as duals: what appears to be computationally easy in one parameterization is involved in the other, and vice versa.

This chapter introduces the two basic Gaussian filter algorithms.

- Chapter 3.2 describes the Kalman filter, which implements the Bayes filter using the moments parameterization for a restricted class of problems with linear dynamics and measurement functions.
- The Kalman filter is extended to nonlinear problems in Chapter 3.3, which describes the extended Kalman filter.
- Chapter 3.4 describes a different nonlinear Kalman filter, known as unscented Kalman filter.
- Chapter 3.5 describes the information filter, which is the dual of the Kalman filter using the canonical parameterization of Gaussians.

3.2 The Kalman Filter

3.2.1 Linear Gaussian Systems

Probably the best studied technique for implementing Bayes filters is the *Kalman filter*, or (*KF*). The Kalman filter was invented by Swerling (1958) and Kalman (1960) as a technique for filtering and prediction in *linear Gaussian systems*, which will be defined in a moment. The Kalman filter implements belief computation for continuous states. It is not applicable to discrete or hybrid state spaces.

The Kalman filter represents beliefs by the moments parameterization: At time t , the belief is represented by the mean μ_t and the covariance Σ_t . Posteriors are *Gaussian* if the following three properties hold, in addition to the Markov assumptions of the Bayes filter.

GAUSSIAN POSTERIOR

1. The state transition probability $p(x_t \mid u_t, x_{t-1})$ must be a *linear* function in its arguments with added Gaussian noise. This is expressed by the following equation:

$$(3.2) \quad x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t$$

Here x_t and x_{t-1} are state vectors, and u_t is the control vector at time t . In our notation, both of these vectors are vertical vectors. They are of the form

$$(3.3) \quad x_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \\ \vdots \\ x_{n,t} \end{pmatrix} \quad \text{and} \quad u_t = \begin{pmatrix} u_{1,t} \\ u_{2,t} \\ \vdots \\ u_{m,t} \end{pmatrix}$$

A_t and B_t are matrices. A_t is a square matrix of size $n \times n$, where n is the dimension of the state vector x_t . B_t is of size $n \times m$, with m being the dimension of the control vector u_t . By multiplying the state and control vector with the matrices A_t and B_t , respectively, the state transition function becomes *linear* in its arguments. Thus, Kalman filters assume linear system dynamics.

The random variable ε_t in (3.2) is a Gaussian random vector that models the uncertainty introduced by the state transition. It is of the same dimension as the state vector. Its mean is zero, and its covariance will be denoted R_t . A state transition probability of the form (3.2) is called a *linear Gaussian*, to reflect the fact that it is linear in its arguments with additive Gaussian noise. Technically, one may also include a constant additive term in (3.2), which is here omitted since it plays no role in the material to come.

Equation (3.2) defines the state transition probability $p(x_t \mid u_t, x_{t-1})$. This probability is obtained by plugging Equation (3.2) into the definition of the multivariate normal distribution (3.1). The mean of the posterior state is given by $A_t x_{t-1} + B_t u_t$ and the covariance by R_t :

$$(3.4) \quad p(x_t \mid u_t, x_{t-1}) = \det(2\pi R_t)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) \right\}$$

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1:      Algorithm Kalman_filter( $\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$ ):
2:           $\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$ 
3:           $\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t$ 
4:           $K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}$ 
5:           $\mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$ 
6:           $\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$ 
7:          return  $\mu_t, \Sigma_t$ 

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Table 3.1 The Kalman filter algorithm for linear Gaussian state transitions and measurements.

2. The measurement probability $p(z_t \mid x_t)$ must also be *linear* in its arguments, with added Gaussian noise:

$$(3.5) \quad z_t = C_t x_t + \delta_t$$

Here C_t is a matrix of size $k \times n$, where k is the dimension of the measurement vector z_t . The vector δ_t describes the measurement noise. The distribution of δ_t is a multivariate Gaussian with zero mean and covariance Q_t . The measurement probability is thus given by the following multivariate normal distribution:

$$(3.6) \quad p(z_t \mid x_t) = \det(2\pi Q_t)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (z_t - C_t x_t)^T Q_t^{-1} (z_t - C_t x_t) \right\}$$

3. Finally, the initial belief $bel(x_0)$ must be normally distributed. We will denote the mean of this belief by μ_0 and the covariance by Σ_0 :

$$(3.7) \quad bel(x_0) = p(x_0) = \det(2\pi \Sigma_0)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (x_0 - \mu_0)^T \Sigma_0^{-1} (x_0 - \mu_0) \right\}$$

These three assumptions are sufficient to ensure that the posterior $bel(x_t)$ is always a Gaussian, for any point in time t . The proof of this non-trivial result can be found below, in the mathematical derivation of the Kalman filter (Chapter 3.2.4).

3.2.2 The Kalman Filter Algorithm

The *Kalman filter algorithm* is depicted in Table 3.1. Kalman filters represent the belief $bel(x_t)$ at time t by the mean μ_t and the covariance Σ_t . The input of the Kalman filter is the belief at time $t - 1$, represented by μ_{t-1} and Σ_{t-1} . To update these parameters, Kalman filters require the control u_t and the measurement z_t . The output is the belief at time t , represented by μ_t and Σ_t .

In lines 2 and 3, the predicted belief $\bar{\mu}$ and $\bar{\Sigma}$ is calculated representing the belief $\bar{bel}(x_t)$ one time step later, but before incorporating the measurement z_t . This belief is obtained by incorporating the control u_t . The mean is updated using the deterministic version of the state transition function (3.2), with the mean μ_{t-1} substituted for the state x_{t-1} . The update of the covariance considers the fact that states depend on previous states through the linear matrix A_t . This matrix is multiplied twice into the covariance, since the covariance is a quadratic matrix.

The belief $\bar{bel}(x_t)$ is subsequently transformed into the desired belief $bel(x_t)$ in lines 4 through 6, by incorporating the measurement z_t . The variable K_t , computed in line 4 is called *Kalman gain*. It specifies the degree to which the measurement is incorporated into the new state estimate, in a way that will become clearer in Chapter 3.2.4. Line 5 manipulates the mean, by adjusting it in proportion to the Kalman gain K_t and the deviation of the actual measurement, z_t , and the measurement predicted according to the measurement probability (3.5). The key concept here is the *innovation*, which is the difference between the actual measurement z_t and the expected measurement $C_t \bar{\mu}_t$ in line 5. Finally, the new covariance of the posterior belief is calculated in line 6, adjusting for the information gain resulting from the measurement.

The Kalman filter is computationally quite efficient. For today's best algorithms, the complexity of matrix inversion is approximately $O(d^{2.4})$ for a matrix of size $d \times d$. Each iteration of the Kalman filter algorithm, as stated here, is lower bounded by (approximately) $O(k^{2.4})$, where k is the dimension of the measurement vector z_t . This (approximate) cubic complexity stems from the matrix inversion in line 4. Even for certain sparse updates discussed in future chapters, it is also at least in $O(n^2)$, where n is the dimension of the state space, due to the multiplication in line 6 (the matrix $K_t C_t$ may be sparse). In many applications—such as the robot mapping applications discussed in later chapters—the measurement space is much lower dimensional than the state space, and the update is dominated by the $O(n^2)$ operations.

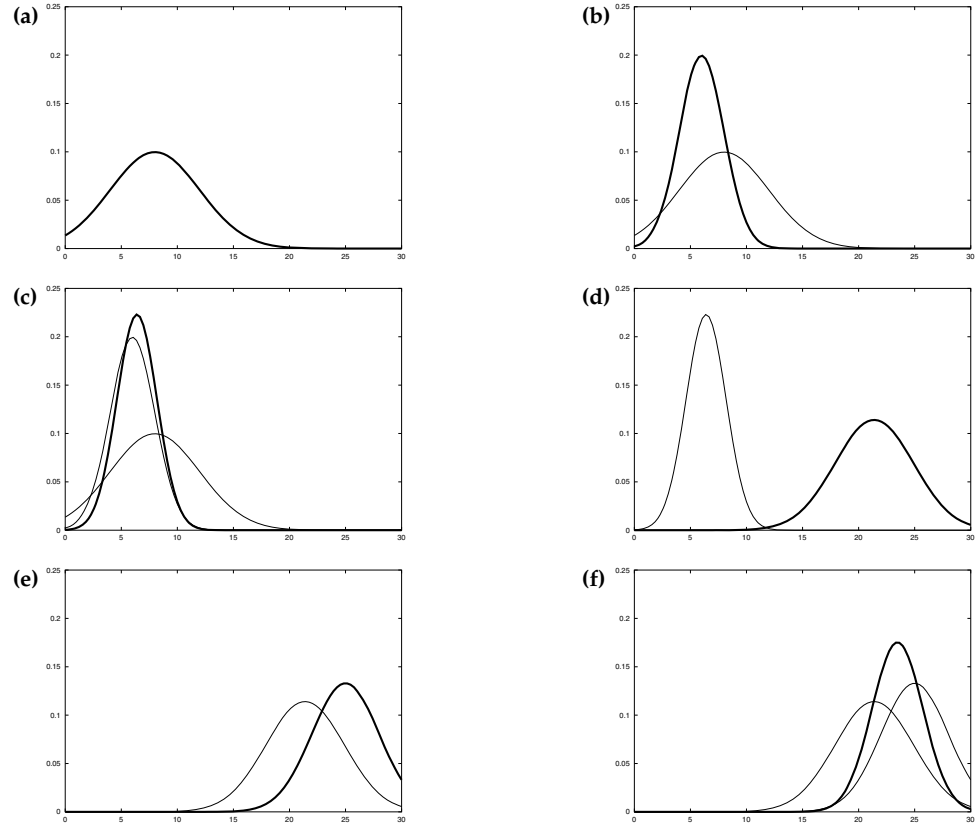


Figure 3.2 Illustration of Kalman filters: (a) initial belief, (b) a measurement (in bold) with the associated uncertainty, (c) belief after integrating the measurement into the belief using the Kalman filter algorithm, (d) belief after motion to the right (which introduces uncertainty), (e) a new measurement with associated uncertainty, and (f) the resulting belief.

3.2.3 Illustration

Figure 3.2 illustrates the Kalman filter algorithm for a simplistic one-dimensional localization scenario. Suppose the robot moves along the horizontal axis in each diagram in Figure 3.2. Let the prior over the robot location be given by the normal distribution shown in Figure 3.2a. The robot queries its sensors on its location (e.g., a GPS system), and those return a measurement that is centered at the peak of the bold Gaussian in Figure 3.2b. This bold Gaussian illustrates this measurement: Its peak is the value predicted

by the sensors, and its width (variance) corresponds to the uncertainty in the measurement. Combining the prior with the measurement, via lines 4 through 6 of the Kalman filter algorithm in Table 3.1, yields the bold Gaussian in Figure 3.2c. This belief's mean lies between the two original means, and its uncertainty radius is smaller than both contributing Gaussians. The fact that the residual uncertainty is smaller than the contributing Gaussians may appear counter-intuitive, but it is a general characteristic of information integration in Kalman filters.

Next, assume the robot moves towards the right. Its uncertainty grows due to the fact that the state transition is stochastic. Lines 2 and 3 of the Kalman filter provide us with the Gaussian shown in bold in Figure 3.2d. This Gaussian is shifted by the amount the robot moved, and it is also wider for the reasons just explained. The robot receives a second measurement illustrated by the bold Gaussian in Figure 3.2e, which leads to the posterior shown in bold in Figure 3.2f.

As this example illustrates, the Kalman filter alternates a *measurement update step* (lines 5-7), in which sensor data is integrated into the present belief, with a *prediction step* (or control update step), which modifies the belief in accordance to an action. The update step decreases and the prediction step increases uncertainty in the robot's belief.

3.2.4 Mathematical Derivation of the KF

This section derives the Kalman filter algorithm in Table 3.1. The section can safely be skipped at first reading; it is only included for completeness.

Up front, the derivation of the KF is largely an exercise in manipulating quadratic expressions. When multiplying two Gaussians, for example, the exponents add. Since both original exponents are quadratic, so is the resulting sum. The remaining exercise is then to come up with a factorization of the result into a form that makes it possible to read off the desired parameters.

Part 1: Prediction

Our derivation begins with lines 2 and 3 of the algorithm, in which the belief $\overline{bel}(x_t)$ is calculated from the belief one time step earlier, $bel(x_{t-1})$. Lines 2 and 3 implement the update step described in Equation (2.41), restated here

for the reader's convenience:

$$(3.8) \quad \overline{bel}(x_t) = \int \underbrace{p(x_t | x_{t-1}, u_t)}_{\sim \mathcal{N}(x_t; A_t x_{t-1} + B_t u_t, R_t)} \underbrace{bel(x_{t-1})}_{\sim \mathcal{N}(x_{t-1}; \mu_{t-1}, \Sigma_{t-1})} dx_{t-1}$$

The belief $bel(x_{t-1})$ is represented by the mean μ_{t-1} and the covariance Σ_{t-1} . The state transition probability $p(x_t | x_{t-1}, u_t)$ was given in (3.4) as a normal distribution over x_t with mean $A_t x_{t-1} + B_t u_t$ and covariance R_t . As we shall show now, the outcome of (3.8) is again a Gaussian with mean $\bar{\mu}_t$ and covariance $\bar{\Sigma}_t$ as stated in Table 3.1.

We begin by writing (3.8) in its Gaussian form:

$$(3.9) \quad \begin{aligned} \overline{bel}(x_t) &= \eta \int \exp \left\{ -\frac{1}{2} (x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) \right\} \\ &\quad \exp \left\{ -\frac{1}{2} (x_{t-1} - \mu_{t-1})^T \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1}) \right\} dx_{t-1} \end{aligned}$$

In short, we have

$$(3.10) \quad \overline{bel}(x_t) = \eta \int \exp \{-L_t\} dx_{t-1}$$

with

$$(3.11) \quad \begin{aligned} L_t &= \frac{1}{2} (x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) \\ &\quad + \frac{1}{2} (x_{t-1} - \mu_{t-1})^T \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1}) \end{aligned}$$

Notice that L_t is quadratic in x_{t-1} ; it is also quadratic in x_t .

Expression (3.10) contains an integral. Solving this integral requires us to reorder the terms in this interval, in a way that might appear counterintuitive at first. In particular, we will decompose L_t into two functions, $L_t(x_{t-1}, x_t)$ and $L_t(x_t)$:

$$(3.12) \quad L_t = L_t(x_{t-1}, x_t) + L_t(x_t)$$

This decomposition will simply be the result of reordering the terms in L_t . A key goal of this decomposition step shall be that the variables in L_t are partitioned into two sets, of which only one will depend on the variable x_{t-1} . The other, $L_t(x_t)$, will not depend on x_{t-1} . As a result, we will be able to move the latter variables out of the integral over the variable x_{t-1} .

This is illustrated by the following transformation:

$$(3.13) \quad \overline{bel}(x_t) = \eta \int \exp \{-L_t\} dx_{t-1}$$

$$\begin{aligned}
&= \eta \int \exp \{-L_t(x_{t-1}, x_t) - L_t(x_t)\} dx_{t-1} \\
&= \eta \exp \{-L_t(x_t)\} \int \exp \{-L_t(x_{t-1}, x_t)\} dx_{t-1}
\end{aligned}$$

Of course, there exist many ways to decompose L_t into two sets that would meet this criterion. The key insight is that we will choose $L_t(x_{t-1}, x_t)$ such that the value of the integral in (3.13) does not depend on x_t . If we succeed in defining such a function $L_t(x_{t-1}, x_t)$, the entire integral over $L_t(x_{t-1}, x_t)$ will simply become a constant relative to the problem of estimating the belief distribution over x_t . Constants are usually captured in the normalization constant η , so under our decomposition we will be able to subsume this constant into η (now for a different actual value of η as above):

$$(3.14) \quad \overline{bel}(x_t) = \eta \exp \{-L_t(x_t)\}$$

Thus, our decomposition would make it possible to eliminate the integral from the belief (3.10). The result is just a normalized exponential over a quadratic function, which turns out to be a Gaussian.

Let us now perform this decomposition. We are seeking a function $L_t(x_{t-1}, x_t)$ quadratic in x_{t-1} . (This function will also depend on x_t , but that shall not concern us at this point.) To determine the coefficients of this quadratic, we calculate the first two derivatives of L_t :

$$(3.15) \quad \frac{\partial L_t}{\partial x_{t-1}} = -A_t^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) + \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1})$$

$$(3.16) \quad \frac{\partial^2 L_t}{\partial x_{t-1}^2} = A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1} =: \Psi_t^{-1}$$

Ψ_t defines the curvature of $L_t(x_{t-1}, x_t)$. Setting the first derivative of L_t to 0 gives us the mean:

$$(3.17) \quad A_t^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) = \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1})$$

This expression is now solved for x_{t-1}

$$\begin{aligned}
(3.18) \quad &\iff A_t^T R_t^{-1} (x_t - B_t u_t) - A_t^T R_t^{-1} A_t x_{t-1} = \Sigma_{t-1}^{-1} x_{t-1} - \Sigma_{t-1}^{-1} \mu_{t-1} \\
&\iff A_t^T R_t^{-1} A_t x_{t-1} + \Sigma_{t-1}^{-1} x_{t-1} = A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1} \\
&\iff (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1}) x_{t-1} = A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1} \\
&\iff \Psi_t^{-1} x_{t-1} = A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1} \\
&\iff x_{t-1} = \Psi_t [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]
\end{aligned}$$

Thus, we now have a quadratic function $L_t(x_{t-1}, x_t)$, defined as follows:

$$(3.19) \quad L_t(x_{t-1}, x_t) = \frac{1}{2} (x_{t-1} - \Psi_t [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}])^T \Psi^{-1} (x_{t-1} - \Psi_t [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}])$$

Clearly, this is not the only quadratic function satisfying our decomposition in (3.12). However, $L_t(x_{t-1}, x_t)$ is of the common quadratic form of the negative exponent of a normal distribution. In fact the function

$$(3.20) \quad \det(2\pi\Psi)^{-\frac{1}{2}} \exp\{-L_t(x_{t-1}, x_t)\}$$

is a valid probability density function (PDF) for the variable x_{t-1} . As the reader easily verifies, this function is of the form defined in (3.1). We know from (2.5) that PDFs integrate to 1. Thus, we have

$$(3.21) \quad \int \det(2\pi\Psi)^{-\frac{1}{2}} \exp\{-L_t(x_{t-1}, x_t)\} dx_{t-1} = 1$$

From this it follows that

$$(3.22) \quad \int \exp\{-L_t(x_{t-1}, x_t)\} dx_{t-1} = \det(2\pi\Psi)^{\frac{1}{2}}$$

The important thing to notice is that the value of this integral is *independent* of x_t , our target variable. Thus, for our problem of calculating a distribution over x_t , this integral is constant. Subsuming this constant into the normalizer η , we get the following expression for Equation (3.13):

$$(3.23) \quad \begin{aligned} \overline{bel}(x_t) &= \eta \exp\{-L_t(x_t)\} \int \exp\{-L_t(x_{t-1}, x_t)\} dx_{t-1} \\ &= \eta \exp\{-L_t(x_t)\} \end{aligned}$$

This decomposition establishes the correctness of (3.14). Notice once again that the normalizers η are *not* the same in both lines.

It remains to determine the function $L_t(x_t)$, which is the difference of L_t , defined in (3.11), and $L_t(x_{t-1}, x_t)$, defined in (3.19):

$$(3.24) \quad \begin{aligned} L_t(x_t) &= L_t - L_t(x_{t-1}, x_t) \\ &= \frac{1}{2} (x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) \\ &\quad + \frac{1}{2} (x_{t-1} - \mu_{t-1})^T \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1}) \\ &\quad - \frac{1}{2} (x_{t-1} - \Psi_t [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}])^T \Psi^{-1} \\ &\quad (x_{t-1} - \Psi_t [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]) \end{aligned}$$

Let us quickly verify that $L_t(x_t)$ indeed does not depend on x_{t-1} . To do so, we substitute back $\Psi_t = (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1}$, and multiply out the terms

above. For the reader's convenience, terms that contain x_{t-1} are underlined (doubly if they are quadratic in x_{t-1}).

$$\begin{aligned}
 (3.25) \quad L_t(x_t) = & \frac{\frac{1}{2} x_{t-1}^T A_t^T R_t^{-1} A_t x_{t-1} - x_{t-1}^T A_t^T R_t^{-1} (x_t - B_t u_t)}{\frac{1}{2} (x_t - B_t u_t)^T R_t^{-1} (x_t - B_t u_t)} \\
 & + \frac{\frac{1}{2} x_{t-1}^T \Sigma_{t-1}^{-1} x_{t-1} - x_{t-1}^T \Sigma_{t-1}^{-1} \mu_{t-1}}{\frac{1}{2} \mu_{t-1}^T \Sigma_{t-1}^{-1} \mu_{t-1}} \\
 & - \frac{\frac{1}{2} x_{t-1}^T (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1}) x_{t-1}}{x_{t-1}^T [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]} \\
 & - \frac{\frac{1}{2} [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]^T (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]}{[A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]}
 \end{aligned}$$

It is now easily seen that all terms that contain x_{t-1} cancel out. This should come at no surprise, since it is a consequence of our construction of $L_t(x_{t-1}, x_t)$.

$$\begin{aligned}
 (3.26) \quad L_t(x_t) = & + \frac{1}{2} (x_t - B_t u_t)^T R_t^{-1} (x_t - B_t u_t) + \frac{1}{2} \mu_{t-1}^T \Sigma_{t-1}^{-1} \mu_{t-1} \\
 & - \frac{1}{2} [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]^T (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} \\
 & [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}]
 \end{aligned}$$

Furthermore, $L_t(x_t)$ is quadratic in x_t . This observation means that $\overline{bel}(x_t)$ is indeed normal distributed. The mean and covariance of this distribution are of course the minimum and curvature of $L_t(x_t)$, which we now easily obtain by computing the first and second derivatives of $L_t(x_t)$ with respect to x_t :

$$\begin{aligned}
 (3.27) \quad \frac{\partial L_t(x_t)}{\partial x_t} = & R_t^{-1} (x_t - B_t u_t) - R_t^{-1} A_t (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} \\
 & [A_t^T R_t^{-1} (x_t - B_t u_t) + \Sigma_{t-1}^{-1} \mu_{t-1}] \\
 = & [R_t^{-1} - R_t^{-1} A_t (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} A_t^T R_t^{-1}] (x_t - B_t u_t) \\
 & - R_t^{-1} A_t (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} \Sigma_{t-1}^{-1} \mu_{t-1}
 \end{aligned}$$

The *inversion lemma* stated (and shown) in Table 3.2 allows us to express the first factor as follows:

$$(3.28) \quad R_t^{-1} - R_t^{-1} A_t (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} A_t^T R_t^{-1} = (R_t + A_t \Sigma_{t-1} A_t^T)^{-1}$$

Hence the desired derivative is given by the following expression:

$$\begin{aligned}
 (3.29) \quad \frac{\partial L_t(x_t)}{\partial x_t} = & (R_t + A_t \Sigma_{t-1} A_t^T)^{-1} (x_t - B_t u_t) \\
 & - R_t^{-1} A_t (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} \Sigma_{t-1}^{-1} \mu_{t-1}
 \end{aligned}$$

Inversion Lemma. For any invertible quadratic matrices R and Q and any matrix P with appropriate dimensions, the following holds true

$$(R + P Q P^T)^{-1} = R^{-1} - R^{-1} P (Q^{-1} + P^T R^{-1} P)^{-1} P^T R^{-1}$$

assuming that all above matrices can be inverted as stated.

Proof. Define $\Psi = (Q^{-1} + P^T R^{-1} P)^{-1}$. It suffices to show that

$$(R^{-1} - R^{-1} P \Psi P^T R^{-1}) (R + P Q P^T) = I$$

This is shown through a series of transformations:

$$\begin{aligned} &= \underbrace{R^{-1} R}_{=I} + R^{-1} P Q P^T - R^{-1} P \Psi P^T \underbrace{R^{-1} R}_{=I} \\ &\quad - R^{-1} P \Psi P^T R^{-1} P Q P^T \\ &= I + R^{-1} P Q P^T - R^{-1} P \Psi P^T - R^{-1} P \Psi P^T R^{-1} P Q P^T \\ &= I + R^{-1} P [Q P^T - \Psi P^T - \Psi P^T R^{-1} P Q P^T] \\ &= I + R^{-1} P [Q P^T - \underbrace{\Psi Q^{-1} Q}_{=I} P^T - \Psi P^T R^{-1} P Q P^T] \\ &= I + R^{-1} P [Q P^T - \underbrace{\Psi \Psi^{-1}}_{=I} Q P^T] \\ &= I + R^{-1} P \underbrace{[Q P^T - Q P^T]}_{=0} = I \end{aligned}$$

Table 3.2 The (specialized) inversion lemma, sometimes called the *Sherman/Morrison formula*.

The minimum of $L_t(x_t)$ is attained when the first derivative is zero.

$$\begin{aligned} (3.30) \quad & (R_t + A_t \Sigma_{t-1} A_t^T)^{-1} (x_t - B_t u_t) \\ &= R_t^{-1} A_t (A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} \Sigma_{t-1}^{-1} \mu_{t-1} \end{aligned}$$

Solving this for the target variable x_t gives us the surprisingly compact result

$$\begin{aligned} (3.31) \quad x_t &= B_t u_t + \underbrace{(R_t + A_t \Sigma_{t-1} A_t^T) R_t^{-1} A_t}_{A_t + A_t \Sigma_{t-1} A_t^T R_t^{-1} A_t} \underbrace{(A_t^T R_t^{-1} A_t + \Sigma_{t-1}^{-1})^{-1} \Sigma_{t-1}^{-1}}_{(\Sigma_{t-1} A_t^T R_t^{-1} A_t + I)^{-1}} \mu_{t-1} \\ &= B_t u_t + A_t \underbrace{(I + \Sigma_{t-1} A_t^T R_t^{-1} A_t) (\Sigma_{t-1} A_t^T R_t^{-1} A_t + I)^{-1}}_{=I} \mu_{t-1} \end{aligned}$$

$$= B_t u_t + A_t \mu_{t-1}$$

Thus, the mean of the belief $\overline{bel}(x_t)$ after incorporating the motion command u_t is $B_t u_t + A_t \mu_{t-1}$. This proves the correctness of line 2 of the Kalman filter algorithm in Table 3.1.

Line 3 is now obtained by calculating the second derivative of $L_t(x_t)$:

$$(3.32) \quad \frac{\partial^2 L_t(x_t)}{\partial x_t^2} = (A_t \Sigma_{t-1} A_t^T + R_t)^{-1}$$

This is the curvature of the quadratic function $L_t(x_t)$, whose inverse is the covariance of the belief $\overline{bel}(x_t)$.

To summarize, we showed that the prediction steps in lines 2 and 3 of the Kalman filter algorithm indeed implement the Bayes filter prediction step. To do so, we first decomposed the exponent of the belief $\overline{bel}(x_t)$ into two functions, $L_t(x_{t-1}, x_t)$ and $L_t(x_t)$. Then we showed that $L_t(x_{t-1}, x_t)$ changes the predicted belief $\overline{bel}(x_t)$ only by a constant factor, which can be subsumed into the normalizing constant η . Finally, we determined the function $L_t(x_t)$ and showed that it results in the mean $\bar{\mu}_t$ and covariance $\bar{\Sigma}_t$ of the Kalman filter prediction $\overline{bel}(x_t)$.

Part 2: Measurement Update

We will now derive the measurement update in lines 4, 5, and 6 (Table 3.1) of our Kalman filter algorithm. We begin with the general Bayes filter mechanism for incorporating measurements, stated in Equation (2.38) and restated here in annotated form:

$$(3.33) \quad bel(x_t) = \eta \underbrace{p(z_t | x_t)}_{\sim \mathcal{N}(z_t; C_t x_t, Q_t)} \underbrace{\overline{bel}(x_t)}_{\sim \mathcal{N}(x_t; \bar{\mu}_t, \bar{\Sigma}_t)}$$

The mean and covariance of $\overline{bel}(x_t)$ are obviously given by $\bar{\mu}_t$ and $\bar{\Sigma}_t$. The measurement probability $p(z_t | x_t)$ was defined in (3.6) to be normal as well, with mean $C_t x_t$ and covariance Q_t . Thus, the product is given by an exponential

$$(3.34) \quad bel(x_t) = \eta \exp \{-J_t\}$$

with

$$(3.35) \quad J_t = \frac{1}{2} (z_t - C_t x_t)^T Q_t^{-1} (z_t - C_t x_t) + \frac{1}{2} (x_t - \bar{\mu}_t)^T \bar{\Sigma}_t^{-1} (x_t - \bar{\mu}_t)$$

This function is quadratic in x_t , hence $bel(x_t)$ is a Gaussian. To calculate its parameters, we once again calculate the first two derivatives of J_t with respect to x_t :

$$(3.36) \quad \frac{\partial J}{\partial x_t} = -C_t^T Q_t^{-1} (z_t - C_t x_t) + \bar{\Sigma}_t^{-1} (x_t - \bar{\mu}_t)$$

$$(3.37) \quad \frac{\partial^2 J}{\partial x_t^2} = C_t^T Q_t^{-1} C_t + \bar{\Sigma}_t^{-1}$$

The second term is the inverse of the covariance of $bel(x_t)$:

$$(3.38) \quad \Sigma_t = (C_t^T Q_t^{-1} C_t + \bar{\Sigma}_t^{-1})^{-1}$$

The mean of $bel(x_t)$ is the minimum of this quadratic function, which we now calculate by setting the first derivative of J_t to zero (and substituting μ_t for x_t):

$$(3.39) \quad C_t^T Q_t^{-1} (z_t - C_t \mu_t) = \bar{\Sigma}_t^{-1} (\mu_t - \bar{\mu}_t)$$

The expression on the left of the equal sign can be transformed as follows:

$$(3.40) \quad \begin{aligned} C_t^T Q_t^{-1} (z_t - C_t \mu_t) &= C_t^T Q_t^{-1} (z_t - C_t \mu_t + C_t \bar{\mu}_t - C_t \bar{\mu}_t) \\ &= C_t^T Q_t^{-1} (z_t - C_t \bar{\mu}_t) - C_t^T Q_t^{-1} C_t (\mu_t - \bar{\mu}_t) \end{aligned}$$

Substituting this back into (3.39) gives us

$$(3.41) \quad C_t^T Q_t^{-1} (z_t - C_t \bar{\mu}_t) = \underbrace{(C_t^T Q_t^{-1} C_t + \bar{\Sigma}_t^{-1})}_{=\Sigma_t^{-1}} (\mu_t - \bar{\mu}_t)$$

and hence we have

$$(3.42) \quad \Sigma_t C_t^T Q_t^{-1} (z_t - C_t \bar{\mu}_t) = \mu_t - \bar{\mu}_t$$

We now define the *Kalman gain* as

$$(3.43) \quad K_t = \Sigma_t C_t^T Q_t^{-1}$$

and obtain

$$(3.44) \quad \mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$$

This proves the correctness of line 5 in the Kalman filter algorithm in Table 3.1.

The Kalman gain, as defined in (3.43), is a function of Σ_t . This is at odds with the fact that we utilize K_t to calculate Σ_t in line 6 of the algorithm. The following transformation shows us how to express K_t in terms of covariances other than Σ_t . It begins with the definition of K_t in (3.43):

$$\begin{aligned}
 (3.45) \quad K_t &= \Sigma_t C_t^T Q_t^{-1} \\
 &= \Sigma_t C_t^T Q_t^{-1} \underbrace{(C_t \bar{\Sigma}_t C_t^T + Q_t) (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}}_{=I} \\
 &= \Sigma_t (C_t^T Q_t^{-1} C_t \bar{\Sigma}_t C_t^T + C_t^T \underbrace{Q_t^{-1} Q_t}_{=I}) (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \\
 &= \Sigma_t (C_t^T Q_t^{-1} C_t \bar{\Sigma}_t C_t^T + C_t^T) (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \\
 &= \Sigma_t (C_t^T Q_t^{-1} C_t \bar{\Sigma}_t C_t^T + \underbrace{\bar{\Sigma}_t^{-1} \bar{\Sigma}_t}_{=I} C_t^T) (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \\
 &= \Sigma_t \underbrace{(C_t^T Q_t^{-1} C_t + \bar{\Sigma}_t^{-1})}_{=\Sigma_t^{-1}} \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \\
 &= \underbrace{\Sigma_t \Sigma_t^{-1}}_{=I} \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \\
 &= \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}
 \end{aligned}$$

This expression proves the correctness of line 4 of our Kalman filter algorithm.

Line 6 is obtained by expressing the covariance using the Kalman gain K_t . The advantage of the calculation in Table 3.1 over the definition in Equation (3.38) lies in the fact that we can avoid inverting the state covariance matrix. This is essential for applications of Kalman filters to high-dimensional state spaces.

Our transformation is once again carried out using the *inversion lemma*, which was already stated in Table 3.2. Here we restate it using the notation of Equation (3.38):

$$(3.46) \quad (\bar{\Sigma}_t^{-1} + C_t^T Q_t^{-1} C_t)^{-1} = \bar{\Sigma}_t - \bar{\Sigma}_t C_t^T (Q_t + C_t \bar{\Sigma}_t C_t^T)^{-1} C_t \bar{\Sigma}_t$$

This lets us arrive at the following expression for the covariance:

$$\begin{aligned}
 (3.47) \quad \Sigma_t &= (C_t^T Q_t^{-1} C_t + \bar{\Sigma}_t^{-1})^{-1} \\
 &= \bar{\Sigma}_t - \bar{\Sigma}_t C_t^T (Q_t + C_t \bar{\Sigma}_t C_t^T)^{-1} C_t \bar{\Sigma}_t \\
 &= [I - \underbrace{\bar{\Sigma}_t C_t^T (Q_t + C_t \bar{\Sigma}_t C_t^T)^{-1} C_t}_{=K_t, \text{ see Eq. (3.45)}}] \bar{\Sigma}_t \\
 &= (I - K_t C_t) \bar{\Sigma}_t
 \end{aligned}$$

This completes our correctness proof, in that it shows the correctness of line 6 of our Kalman filter algorithm.

3.3 The Extended Kalman Filter

3.3.1 Why Linearize?

The assumptions that observations are linear functions of the state and that the next state is a linear function of the previous state are crucial for the correctness of the Kalman filter. The observation that any linear transformation of a Gaussian random variable results in another Gaussian random variable played an important role in the derivation of the Kalman filter algorithm. The efficiency of the Kalman filter is then due to the fact that the parameters of the resulting Gaussian can be computed in closed form.

Throughout this and the following chapters, we will illustrate properties of different density representations using the transformation of a one-dimensional Gaussian random variable. Figure 3.3a illustrates the *linear* transformation of such a random variable. The graph on the lower right shows the density of the random variable $X \sim \mathcal{N}(x; \mu, \sigma^2)$. Let us assume that X is passed through the linear function $y = ax + b$, shown in the upper right graph. The resulting random variable, Y , is distributed according to a Gaussian with mean $a\mu + b$ and variance $a^2\sigma^2$. This Gaussian is illustrated by the gray area in the upper left graph of Figure 3.3a. The reader may notice that this example is closely related to the next state update of the Kalman filter, with $X = x_{t-1}$ and $Y = x_t$ but without an additive noise variable; see also Equation (3.2).

Unfortunately, state transitions and measurements are rarely linear in practice. For example, a robot that moves with constant translational and rotational velocity typically moves on a circular trajectory, which cannot be described by linear state transitions. This observation, along with the assumption of unimodal beliefs, renders plain Kalman filters, as discussed so far, inapplicable to all but the most trivial robotics problems.

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The *extended Kalman filter*, or *EKF*, relaxes one of these assumptions: the linearity assumption. Here the assumption is that the state transition probability and the measurement probabilities are governed by *nonlinear* functions g and h , respectively:

$$(3.48) \quad x_t = g(u_t, x_{t-1}) + \varepsilon_t$$

$$(3.49) \quad z_t = h(x_t) + \delta_t$$